TLS Implementation with RSA, ECDHE & AES

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Information Security – Assignment 1
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https://github.com/Akshaymalwey/TLS-1.3-from-scratch

Project Workflow Summary

1. **Key & Certificate Setup**
   - Generated RSA public/private key pairs using OpenSSL
   - Created three entities: Client, Server, and self-signed Root CA
   - Root CA issued and signed certificates for both Client and Server
2. **TLS-like Handshake**
   - Client sends **ClientHello** containing:
     - Supported curve (prime256v1)
     - Client random value
     - Client ephemeral public key (ECDHE)
   - Server responds with **ServerHello** containing:
     - Server random value
     - Server certificate (signed by Root CA)
     - Server ephemeral public key
     - RSA signature over handshake parameters
   - Client verifies certificate chain and signature authenticity
3. **Key Exchange & Secure Communication**
   - Both sides compute shared secret using ECDHE
   - Session keys derived using HKDF with SHA-256
   - Secure communication established using AES-256-GCM

Code Implementation Snippets

```c
//hkdfc key generation
const sharedSecret = ecdh.computeSecret(clientEphemeralPk)
function deriveSessionKeys(sharedSecret, clientRandom, serverRandom){
    const salt = Buffer.concat([clientRandom, serverRandom])
    const keyMat = crypto.hkdfSync(
            'sha256',
            sharedSecret,
            salt,
            Buffer.from('handshake data'),
            64
        )

    return {
        clientWKey: keyMat.slice(0,32),
        serverWKey: keyMat.slice(32,64)
    }
}
```

```c
//function to encrypting the message
function encryptMessage(plaintext, key){
    const iv = crypto.randomBytes(12)
    const cipher = crypto.createCipheriv('aes-256-gcm', key, iv)
    const encrypted = Buffer.concat([
        cipher.update(plaintext, 'utf8'),
        cipher.final()
    ])
    const authTag = cipher.getAuthTag();

    return {
        iv: iv.toString('base64'),
        ciphertext: encrypted.toString('base64'),
        tag: authTag.toString('base64')
    }

}
```

```c
//sign the ecdh public key by pvt RSA key of server
const sign = crypto.createSign('sha256')
sign.update(Buffer.concat([clientRandom, serverRandom, serverEphemeralPk]))
sign.end()
const signature = sign.sign(serverKey)
```


```
-----BEGIN CERTIFICATE-----
MIIDazCCAlOgAwIBAgIUJmy43jevrFnGYgTNhw1P2r75a6YwDQYJKoZIhvcNAQEL
BQAwRTELMAkGA1UEBhMCaW4xCzAJBgNVBAgMAmptMQswCQYDVQQKDAJpajELMAkG
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qUagz6haLxwW4yCcZHIy
-----END CERTIFICATE-----
```
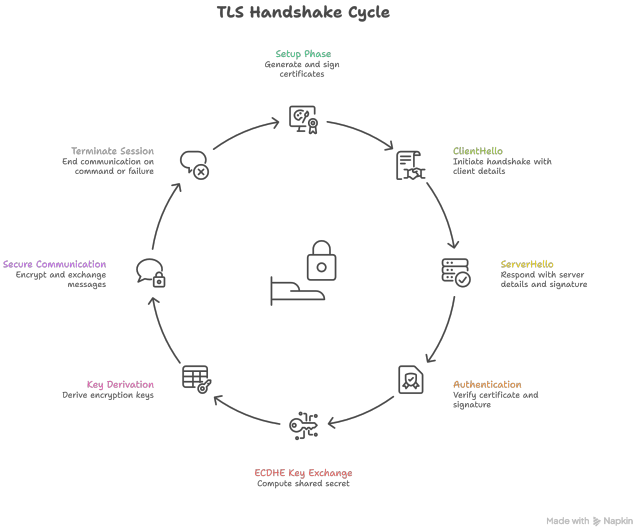

Complete Workflow



Thank You